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Course : MATH6030001 - Linear Algebra
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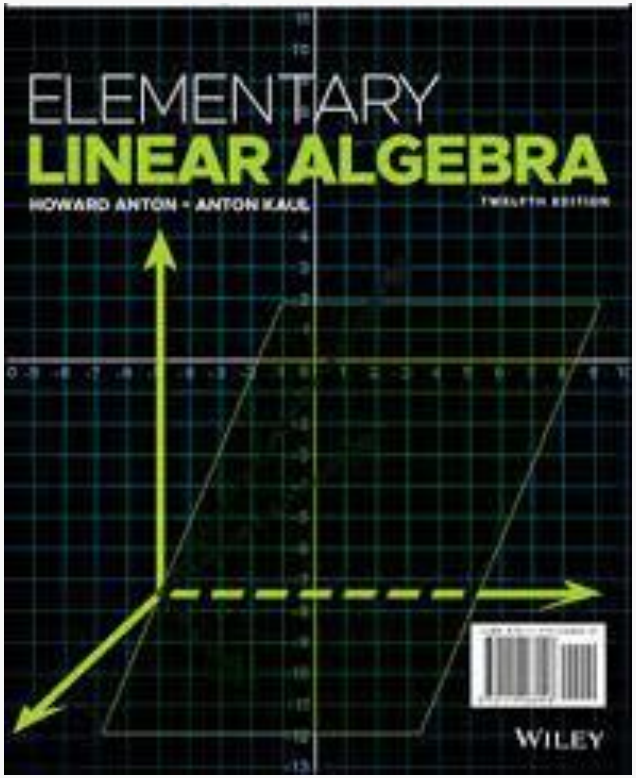
Matrix

Session 03-06

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Acknowledgement



These slides have been adapted from:

Howard Anton and Anton Kaul. *Elementary linear algebra: Applications version (12th Edition)*. John Wiley & Sons, 2019.

Section 1.3-1.7, Chapter 2

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Learning Objectives

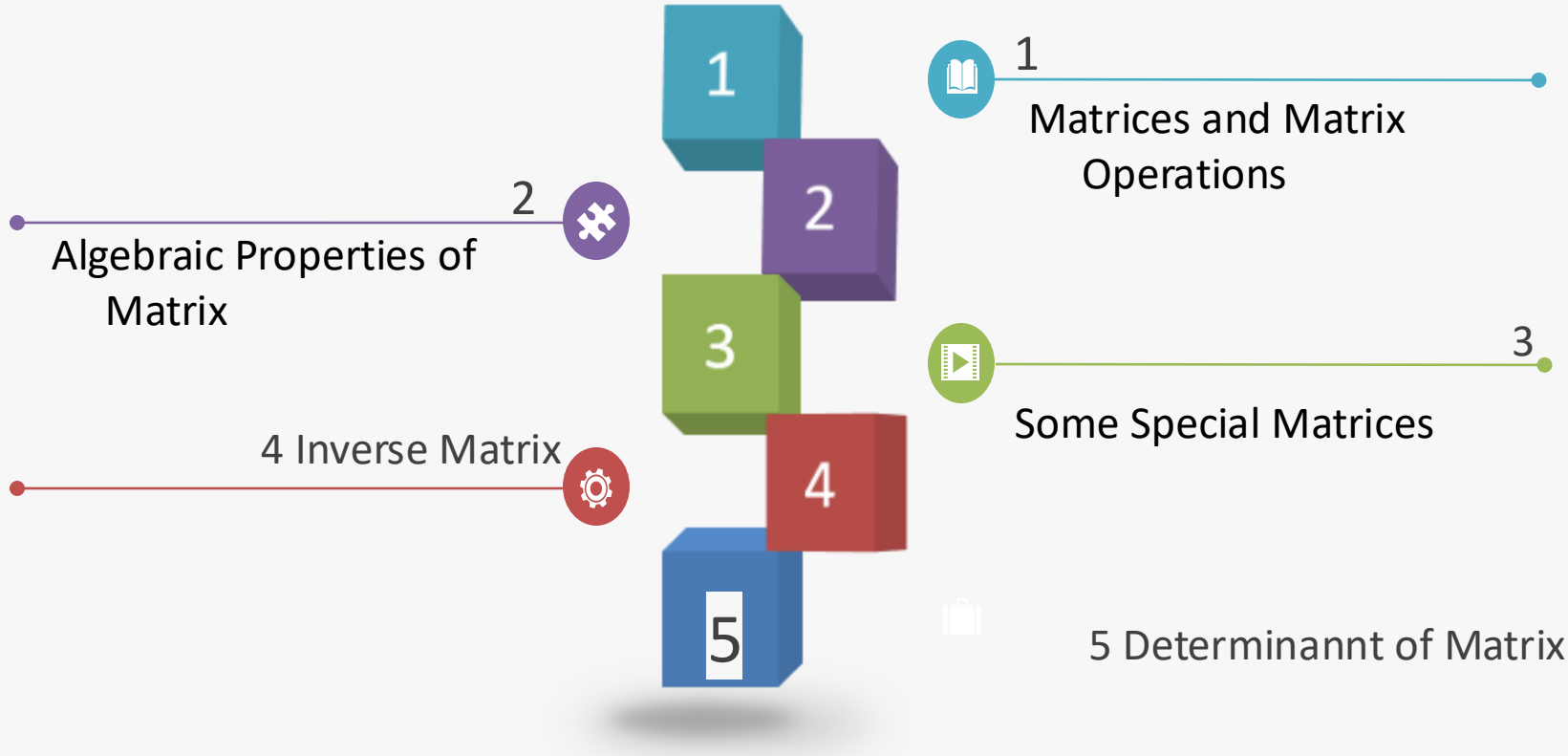
LO 2: Evaluate determinant and inverse of matrix and their applications.



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Contents



Matrices and Matrix Operations

Basic Terminologies (1/4)

Definition

A ***matrix*** is a rectangular array of numbers. The numbers in the array are called the ***entries*** in the matrix.

Examples

Some examples of matrices are

$$\begin{bmatrix} 1 & 2 \\ 3 & 0 \\ -1 & 4 \end{bmatrix}, \quad [2 \quad 1 \quad 0 \quad -3], \quad \begin{bmatrix} e & \pi & -\sqrt{2} \\ 0 & \frac{1}{2} & 1 \\ 0 & 0 & 0 \end{bmatrix}, \quad \begin{bmatrix} 1 \\ 3 \end{bmatrix}, \quad [4]$$

The ***size*** of a matrix is described in terms of the number of rows (horizontal lines) and columns (vertical lines) it contains. For example, the first matrix in above example has three rows and two columns, so its size is 3 by 2 (written 3x2).

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Basic Terminologies (2/4)

In a size description, the first number always denotes the number of rows, and the second denotes the number of columns. The remaining matrices in above example have sizes 1x4, 3x3, 2x1, and 1x1, respectively.

A matrix with only one column is called a ***column vector*** or a ***column matrix***, and a matrix with only one row is called a ***row vector*** or a ***row matrix***. In previous example, the 2x1 matrix is a column vector, the 1x4 matrix is a row vector, and the 1x1 matrix is both a row vector and a column vector.

When discussing matrices, it is common to refer to numerical quantities as ***scalars***. Unless stated otherwise, *scalars will be real numbers.*

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Basic Terminologies (3/4)

The entry that occurs in row *i* and column *j* of a matrix *A* will be denoted by a_{ij}. Thus a general *m* x *n* matrix as :

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}$$

When a compact notation is desired, the preceding matrix can be written as :

$$[a_{ij}]_{m \times n} \text{ or } [a_{ij}]$$

the first notation being used when it is important in the discussion to know the size, and the second being used when the size need not be emphasized.

Usually, we will match the letter denoting a matrix with the letter denoting its entries. The entry in row *i* and column *j* of a matrix *A* is also commonly denoted by the symbol $(A)_{ij}$ or a_{ij}.

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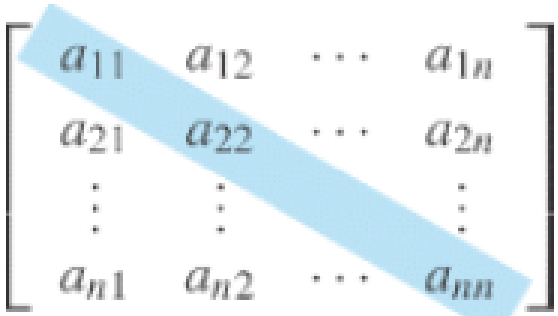
Basic Terminologies (4/4)

As an example, in the matrix

$$A = \begin{bmatrix} 2 & -3 \\ 7 & 0 \end{bmatrix}$$

we have $(A)_{11} = 2$, $(A)_{12} = -3$, $(A)_{21} = 7$ and $(A)_{22} = 0$.

A matrix *A* with *n* rows and *n* columns is called a ***square matrix of order n***, and the shaded entries $a_{11}, a_{22}, \ldots, a_{nn}$ in the matrix below are said to be on the ***main diagonal*** of *A*.



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Equality of Matrices

Definition

Two matrices are defined to be ***equal*** if they have the same size and their corresponding entries are equal.

The equality of two matrices

$$A=[a_{ij}] \text{ and } B=[b_{ij}]$$

of the same size can be expressed by writing $a_{ij}=b_{ij}$, for all values of i and j in those matrices.

Example

Consider the matrices

$$A=\begin{bmatrix}2 & 1\\ 3 & x\end{bmatrix},\quad B=\begin{bmatrix}2 & 1\\ 3 & 5\end{bmatrix},\quad C=\begin{bmatrix}2 & 1 & 0\\ 3 & 4 & 0\end{bmatrix}$$

If x=5 then A=B, but matrices *A* and *B* are not equal, since not all of their corresponding entries are equal. There is no value of *x* for which A=C since *A* and *C* have different sizes.

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Matrix Addition and Substraction

Definition

If *A* and *B* are matrices of the same size, then the ***sum A+B*** is the matrix obtained by adding the entries of *B* to the corresponding entries of *A*, and the ***difference A-B*** is the matrix obtained by subtracting the entries of *B* from the corresponding entries of *A*. Matrices of different sizes cannot be added or subtracted.

In matrix notation, if $A=[a_{ij}]$ and $B=[b_{ij}]$ have the same size, then

$$(A+B)_{ij} = (A)_{ij}+(B)_{ij} = a_{ij}+b_{ij}$$

and

$$(A-B)_{ij} = (A)_{ij}-(B)_{ij} = a_{ij}-b_{ij}$$

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Example

Consider the matrices

$$A=\begin{bmatrix} 2 & 1 & 0 & 3 \\ -1 & 0 & 2 & 4 \\ 4 & -2 & 7 & 0 \end{bmatrix}, \quad B=\begin{bmatrix} -4 & 3 & 5 & 1 \\ 2 & 2 & 0 & -1 \\ 3 & 2 & -4 & 5 \end{bmatrix}, \quad C=\begin{bmatrix} 1 & 1 \\ 2 & 2 \end{bmatrix}$$

Then

$$A+B=\begin{bmatrix} -2 & 4 & 5 & 4 \\ 1 & 2 & 2 & 3 \\ 7 & 0 & 3 & 5 \end{bmatrix} \quad \text{and} \quad A-B=\begin{bmatrix} 6 & -2 & -5 & 2 \\ -3 & -2 & 2 & 5 \\ 1 & -4 & 11 & -5 \end{bmatrix}$$

The expressions $A+C$, $B+C$, $A-C$, and $B-C$ are undefined.

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Scalar Multiplication

Definition

If *A* is any matrix and *c* is any scalar, then the ***product*** *c.A* is the matrix obtained by multiplying each entry of the matrix *A* by *c*. The matrix *c.A* is said to be a ***scalar multiple*** of *A*.

In matrix notation, if $A=[a_{ij}]$, then

$$(c.A)_{ij} = c.(A)_{ij} = c.a_{ij}$$

Example

For the matrices

$$A=\begin{bmatrix}2 & 3 & 4\\ 1 & 3 & 1\end{bmatrix},\quad B=\begin{bmatrix}0 & 2 & 7\\ -1 & 3 & -5\end{bmatrix},\quad C=\begin{bmatrix}9 & -6 & 3\\ 3 & 0 & 12\end{bmatrix}$$

we have

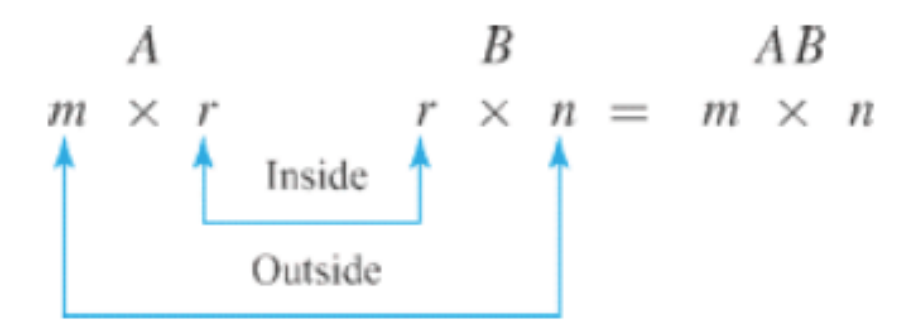
$$2A=\begin{bmatrix}4 & 6 & 8\\ 2 & 6 & 2\end{bmatrix},\quad (-1)B=\begin{bmatrix}0 & -2 & -7\\ 1 & -3 & 5\end{bmatrix},\quad \frac{1}{3}C=\begin{bmatrix}3 & -2 & 1\\ 1 & 0 & 4\end{bmatrix}$$

It is common practice to denote $(-1)B$ by $-B$.

Matrix Multiplication

Definition

If A is m x r an matrix and B is an r x n matrix, then the ***product*** AB is the m x n matrix whose entries are determined as follows: To find the entry in row i and column j of AB, single out row i from the matrix A and column j from the matrix B. Multiply the corresponding entries from the row and column together, and then add up the resulting products.



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Example

Consider the matrices

$$A=\begin{bmatrix}1 & 2 & 4\\ 2 & 6 & 0\end{bmatrix},\quad B=\begin{bmatrix}4 & 1 & 4 & 3\\ 0 & -1 & 3 & 1\\ 2 & 7 & 5 & 2\end{bmatrix}$$

Since A is a 2×3 matrix and B is a 3×4 matrix, the product AB is a 2×4 matrix. To determine, for example, the entry in row 2 and column 3 of AB, we single out row 2 from A and column 3 from B. Then, as illustrated below, we multiply corresponding entries together and add up these products.

$$\begin{bmatrix}1 & 2 & 4\\ 2 & 6 & 0\end{bmatrix}\begin{bmatrix}4 & 1 & 4 & 3\\ 0 & -1 & 3 & 1\\ 2 & 7 & 5 & 2\end{bmatrix}=\begin{bmatrix}\square & \square & \square & \square\\ \square & \square & \boxed{26} & \square\end{bmatrix}$$

$$(2\cdot 4)+(6\cdot 3)+(0\cdot 5)=26$$

The entry in row 1 and column 4 of AB is computed as follows:

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Example (Cont.)

$$\begin{bmatrix} 1 & 2 & 4 \\ 2 & 6 & 0 \end{bmatrix}\begin{bmatrix} 4 & 1 & 4 & 3 \\ 0 & -1 & 3 & 1 \\ 2 & 7 & 5 & 2 \end{bmatrix} = \begin{bmatrix} \square & \square & \square & \boxed{13} \\ \square & \square & \square & \square \end{bmatrix}$$

$$(1 \cdot 3) + (2 \cdot 1) + (4 \cdot 2) = 13$$

The computations for the remaining entries are

$$(1.4) + (2.0) + (4.2) = 12$$
$$(1.1) - (2.1) + (4.7) = 27$$
$$(1.4) + (2.3) + (4.5) = 30$$
$$(2.4) + (6.0) + (0.2) = 8$$
$$(2.1) - (6.1) + (0.7) = -4$$
$$(2.3) + (6.1) + (0.2) = 12$$

$$AB = \begin{bmatrix} 12 & 27 & 30 & 13 \\ 8 & -4 & 26 & 12 \end{bmatrix}$$

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General Formula of Matrix Multiplication

If A=[a_{ij}] is an *m* x *r* matrix and B=[b_{ij}] is an *r* x *n* matrix, then :

$$AB = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1r} \\ a_{21} & a_{22} & \cdots & a_{2r} \\ \vdots & \vdots & & \vdots \\ a_{i1} & a_{i2} & \cdots & a_{ir} \\ \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mr} \end{bmatrix} \begin{bmatrix} b_{11} & b_{12} & \cdots & b_{ij} & \cdots & b_{1n} \\ b_{21} & b_{22} & \cdots & b_{2j} & \cdots & b_{2n} \\ \vdots & \vdots & & \vdots & & \vdots \\ b_{r1} & b_{r2} & \cdots & b_{rj} & \cdots & b_{rn} \end{bmatrix}$$

the entry (AB)ij in row i and column j of AB is given by

$$(AB)_{ij} = a_{i1}b_{1j} + a_{i2}b_{2j} + a_{i3}b_{3j} + \cdots + a_{ir}b_{rj}$$

Notes : Suppose that *A, B*, and *C* are 3x4, 4x7 and 7x3 matrices, respectively. Then, *AB* is defined and is a 3x7 matrix; *BC* is defined and is a 4x3 matrix; and *CA* is defined and is a 7x4 matrix. The products *AC, CB*, and *BA* are all undefined.

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Matrix Form of a Linear System

Matrix multiplication has an important application to systems of linear equations. Consider a system of *m* linear equations in *n* unknowns:

$$\begin{array}{ccccccc} a_{11}x_1 & + & a_{12}x_2 & + \cdots + & a_{1n}x_n & = & b_1 \\ a_{21}x_1 & + & a_{22}x_2 & + \cdots + & a_{2n}x_n & = & b_2 \\ \vdots & & \vdots & & \vdots & & \vdots \\ a_{m1}x_1 & + & a_{m2}x_2 & + \cdots + & a_{mn}x_n & = & b_m \end{array}$$

Since two matrices are equal if and only if their corresponding entries are equal, we can replace the *m* equations in this system by the single matrix equation

$$\begin{bmatrix} a_{11}x_1 & + & a_{12}x_2 & + \cdots + & a_{1n}x_n \\ a_{21}x_1 & + & a_{22}x_2 & + \cdots + & a_{2n}x_n \\ \vdots & & \vdots & & \vdots \\ a_{m1}x_1 & + & a_{m2}x_2 & + \cdots + & a_{mn}x_n \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}$$

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Matrix Form of a Linear System (Cont.)

The *m* x 1 matrix on the left side of this equation can be written as a product to give

$$\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}$$

If we designate these matrices by *A*, **x**, and **b**, respectively, then we can replace the original system of *m* equations in *n* unknowns has been replaced by the single matrix equation

$$A.\mathbf{x}=\mathbf{b}$$

The matrix *A* in this equation is called the ***coefficient matrix*** of the system. The ***augmented matrix*** for the system is obtained by adjoining **b** to *A* as the last column; thus the augmented matrix is

$$[A|\mathbf{b}] = \left[\begin{array}{cccc|c} a_{11} & a_{12} & \cdots & a_{1n} & b_1 \\ a_{21} & a_{22} & \cdots & a_{2n} & b_2 \\ \vdots & \vdots & & \vdots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} & b_m \end{array}\right]$$

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Transpose of a Matrix

Definition

If A is any m x n matrix, then the ***transpose of A***, denoted by A^T, is defined to be the n x m matrix that results by interchanging the rows and columns of A; that is, the first column of A^T is the first row of A, the second column of A^T is the second row of A, and so forth.

Example

The following are some examples of matrices and their transpo

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \end{bmatrix}, \quad B = \begin{bmatrix} 2 & 3 \\ 1 & 4 \\ 5 & 6 \end{bmatrix}, \quad C = [1 \quad 3 \quad 5], \quad D = [4]$$

$$A^T = \begin{bmatrix} a_{11} & a_{21} & a_{31} \\ a_{12} & a_{22} & a_{32} \\ a_{13} & a_{23} & a_{33} \\ a_{14} & a_{24} & a_{34} \end{bmatrix}, \quad B^T = \begin{bmatrix} 2 & 1 & 5 \\ 3 & 4 & 6 \end{bmatrix}, \quad C^T = \begin{bmatrix} 1 \\ 3 \\ 5 \end{bmatrix}, \quad D^T = [4]$$

Transpose of a Matrix (Cont.)

Observe that not only are the columns of A^T the rows of A, but the rows of A^T are the columns of A. Thus the entry in row i and column j of A^T is the entry in row j and column i of A; that is,

$$(A^T)_{ij} = (A)_{ji}$$

In the special case where A is a square matrix, the transpose of A can be obtained by interchanging entries that are symmetrically positioned about the main diagonal. We see that A^T can also be obtained by "reflecting" A about its main diagonal.

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$$A = \begin{bmatrix} 1 & -2 & 4 \\ 3 & 7 & 0 \\ -5 & 8 & 6 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & -2 & 4 \\ 3 & 7 & 0 \\ -5 & 8 & 6 \end{bmatrix} \rightarrow A^T = \begin{bmatrix} 1 & 3 & -5 \\ -2 & 7 & 8 \\ 4 & 0 & 6 \end{bmatrix}$$

Interchange entries that are symmetrically positioned about the main diagonal.

Properties of the Transpose

Theorem

If the sizes of the matrices are such that the stated operations can be performed, then:

(a) $(A^T)^T = A$

(b) $(A + B)^T = A^T + B^T$

(c) $(A - B)^T = A^T - B^T$

(d) $(k.A)^T = k.A^T$

(e) $(A.B)T = B^T.A^T$ (*The transpose of a product of any number of matrices is the product of the transposes in the reverse order)*

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Trace of a Matrix

Definition

If *A* is a square matrix, then the ***trace of A***, denoted by tr(*A*), is defined to be the sum of the entries on the main diagonal of *A*. The trace of *A* is undefined if *A* is not a square matrix.

Example

The following are examples of matrices and their traces.

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}, \quad B = \begin{bmatrix} -1 & 2 & 7 & 0 \\ 3 & 5 & -8 & 4 \\ 1 & 2 & 7 & -3 \\ 4 & -2 & 1 & 0 \end{bmatrix}$$

$$\text{tr}(A) = a_{11} + a_{22} + a_{33} \quad \text{tr}(B) = -1 + 5 + 7 + 0 = 11$$

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Algebraic Properties of Matrix

Properties of Matrix Arithmetic

Theorem

Assuming that the sizes of the matrices are such that the indicated operations can be performed, the following rules of matrix arithmetic

(a) $A + B = B + A$ **(Commutative law for addition)**

(b) $A + (B + C) = (A + B) + C$ **(Associative law for addition)**

(c) $A(BC) = (AB)C$ **(Associative law for multiplication)**

(d) $A(B + C) = AB + AC$ **(Left distributive law)**

(e) $(B + C)A = BA + CA$ **(Right distributive law)**

(f) $A(B - C) = AB - AC$

(g) $(B - C)A = BA - CA$

(h) $a(B + C) = aB + aC$

(i) $a(B - C) = aB - aC$

(j) $(a + b)C = aC + bC$

(k) $(a - b)C = aC - bC$

(l) $a(bC) = (ab)C$

(m) $a(BC) = (aB)C = B(aC)$

Associativity of Matrix Multiplication

As an illustration of the associative law for matrix multiplication, consider

$$A=\begin{bmatrix}1 & 2\\ 3 & 4\\ 0 & 1\end{bmatrix},\quad B=\begin{bmatrix}4 & 3\\ 2 & 1\end{bmatrix},\quad C=\begin{bmatrix}1 & 0\\ 2 & 3\end{bmatrix}$$

Then

$$AB=\begin{bmatrix}1 & 2\\ 3 & 4\\ 0 & 1\end{bmatrix}\begin{bmatrix}4 & 3\\ 2 & 1\end{bmatrix}=\begin{bmatrix}8 & 5\\ 20 & 13\\ 2 & 1\end{bmatrix}\quad\text{and}\quad BC=\begin{bmatrix}4 & 3\\ 2 & 1\end{bmatrix}\begin{bmatrix}1 & 0\\ 2 & 3\end{bmatrix}=\begin{bmatrix}10 & 9\\ 4 & 3\end{bmatrix}$$

Thus

$$(AB)C=\begin{bmatrix}8 & 5\\ 20 & 13\\ 2 & 1\end{bmatrix}\begin{bmatrix}1 & 0\\ 2 & 3\end{bmatrix}=\begin{bmatrix}18 & 15\\ 46 & 39\\ 4 & 3\end{bmatrix}$$

and

$$A(BC)=\begin{bmatrix}1 & 2\\ 3 & 4\\ 0 & 1\end{bmatrix}\begin{bmatrix}10 & 9\\ 4 & 3\end{bmatrix}=\begin{bmatrix}18 & 15\\ 46 & 39\\ 4 & 3\end{bmatrix}$$

So (AB)C=A(BC), as it is stated in the theorem part (c).

Matrix Multiplication is not Commutative

Do not think that *all* laws of real arithmetic carry over to matrix arithmetic. For example, you know that in real arithmetic it is always true that a.b=b.a, which is called the *commutative law for multiplication*. In matrix arithmetic, however, the equality of *AB* and *BA* can fail for three possible reasons :

1. *AB* may be defined and *BA* may not (for example, if *A* is 2x3 and *B* is 3x4).
2. *AB* and *BA* may both be defined, but they may have different sizes (for example, if *A* is 2x3 and *B* is 3x2).
3. *AB* and *BA* may both be defined and have the same size, but the two matrices may be different (as illustrated in the next example).

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Example

Consider the matrice $A=\begin{bmatrix}-1 & 0\\ 2 & 3\end{bmatrix}$ and $B=\begin{bmatrix}1 & 2\\ 3 & 0\end{bmatrix}$

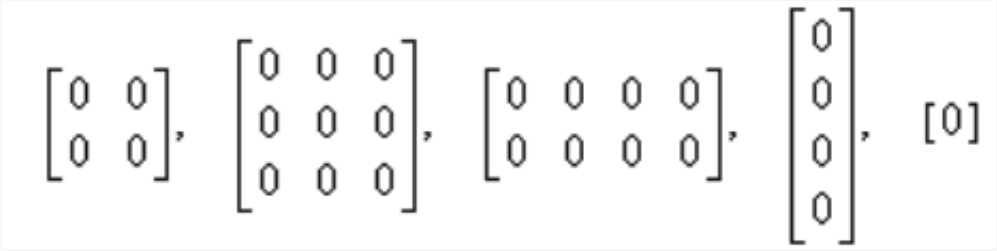
Multiplying gives $AB=\begin{bmatrix}-1 & -2\\ 11 & 4\end{bmatrix}$ and $BA=\begin{bmatrix}3 & 6\\ -3 & 0\end{bmatrix}$

Thus, AB≠BA.

Zero Matrices

Definition

A matrix whose entries are all zero is called a ***zero matrix***. Some examples are



We will denote a zero matrix by *0* unless it is important to specify its size, in which case we will denote the *m* x *n* zero matrix by 0_{mxn}.

It should be evident that if *A* and *0* are matrices with the same size, then A+*0* = *0*+A = A.

Thus, *0* plays the same role in this matrix equation that the number 0 plays in the numerical equation a+0 = 0+a = a.

Properties of Zero Matrices

Theorem

If c is a scalar, and if the sizes of the matrices are such that the operations can be performed, then:

(a) $A + 0 = 0 + A = A$

(b) $A - 0 = A$

(c) $A - A = A + (-A) = 0$

(d) $0.A = 0$

(e) If $c.A = 0$, then $c=0$ or $A=0$

Notes : In real arithmetic, we know that :

- If $a.b = a.c$ and $a \neq 0$, then $b = c$. (The cancellation law)
- If $a.b = 0$, then at least one of the factors on he left is 0.

However, those laws are not universally true in matrix arithmetic.

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Other Laws which Doesn't Hold in Matrix Multiplication

Example 1

Consider the matrices

$$A=\begin{bmatrix}0 & 1\\ 0 & 2\end{bmatrix},\quad B=\begin{bmatrix}1 & 1\\ 3 & 4\end{bmatrix},\quad C=\begin{bmatrix}2 & 5\\ 3 & 4\end{bmatrix}$$

We can compute that

$$AB=AC=\begin{bmatrix}3 & 4\\ 6 & 8\end{bmatrix}$$

Although A≠*0*, canceling *A* from both sides of the equation AB=AC would lead to the incorrect conclusion that B = C. Thus, the cancellation law does not hold, in general, for matrix multiplication.

Example 2

Here are two matri[illegible] t A ≠ *0* and B ≠ *0*.

$$A=\begin{bmatrix}0 & 1\\ 0 & 2\end{bmatrix},\quad B=\begin{bmatrix}3 & 7\\ 0 & 0\end{bmatrix}$$

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Identity Matrices

Definition

A square matrix with 1's on the main diagonal and zeros elsewhere is called an ***identity matrix***. Some examples are

$$[1],\ \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix},\ \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix},\ \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

An ***identity matrix*** is denoted by the letter *I*. If it is important to emphasize the size, we will write I_n for the *n* x *n* identity matrix.

The identity matrices play the same role in these matrix equations that the number 1 plays in the numerical equation : a.1 = 1.a = a, as explained in the next slide.

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Properties of Identity Matrices

To explain the role of identity matrices in matrix arithmetic, let us consider the effect of multiplying a general 2x3 matrix *A* on each side by an identity matrix. Multiplying on the right by the 3x3 identity matrix yields

$$AI_3 = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{bmatrix} = A$$

and multiplying on the left by the 2x2 identity matrix yields

$$I_2 A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{bmatrix} = A$$

The same result holds in general; that is, if *A* is *m* x *n* any matrix, then :

$$A.I_n = A \text{ and } I_m.A = A.$$

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Some Special Matrices

Diagonal Matrices

Definition

A square matrix in which all the entries off the main diagonal are zero is called a *diagonal matrix*. Here are some examples:

$$\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}, \quad \begin{bmatrix} 2 & 0 \\ 0 & -5 \end{bmatrix}, \quad \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad \begin{bmatrix} 6 & 0 & 0 & 0 \\ 0 & -4 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 8 \end{bmatrix}$$

A general *n* x *n* diagonal matrix D can be written as

$$D = \begin{bmatrix} d_1 & 0 & \cdots & 0 \\ 0 & d_2 & \cdots & 0 \\ \vdots & \vdots & & \vdots \\ 0 & 0 & \cdots & d_n \end{bmatrix}$$

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Properties of Diagonal Matrices

If D is the diagonal matrix 1 and k is a positive integer, then

$$D^k = \begin{bmatrix} d_1^k & 0 & \dots & 0 \\ 0 & d_2^k & \dots & 0 \\ \vdots & \vdots & & \vdots \\ 0 & 0 & \dots & d_n^k \end{bmatrix}$$

Example

If $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -3 & 0 \\ 0 & 0 & 2 \end{bmatrix}$, then $A^5 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -243 & 0 \\ 0 & 0 & 32 \end{bmatrix}$.

Note : Power of a square matrix A to the n, denoted by A^n, is defined as : A^n = A.A.....A [n factors]. We will discuss more about power of matrix in next topic.

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Properties of Diagonal Matrices

Matrix products that involve diagonal factors are especially easy to compute. For example,

$$\begin{bmatrix} d_1 & 0 & 0 \\ 0 & d_2 & 0 \\ 0 & 0 & d_3 \end{bmatrix}\begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \end{bmatrix} = \begin{bmatrix} d_1a_{11} & d_1a_{12} & d_1a_{13} & d_1a_{14} \\ d_2a_{21} & d_2a_{22} & d_2a_{23} & d_2a_{24} \\ d_3a_{31} & d_3a_{32} & d_3a_{33} & d_3a_{34} \end{bmatrix}$$

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \\ a_{41} & a_{42} & a_{43} \end{bmatrix}\begin{bmatrix} d_1 & 0 & 0 \\ 0 & d_2 & 0 \\ 0 & 0 & d_3 \end{bmatrix} = \begin{bmatrix} d_1a_{11} & d_2a_{12} & d_3a_{13} \\ d_1a_{21} & d_2a_{22} & d_3a_{23} \\ d_1a_{31} & d_2a_{32} & d_3a_{33} \\ d_1a_{41} & d_2a_{42} & d_3a_{43} \end{bmatrix}$$

In words, to multiply a matrix A on the left by a diagonal matrix D, one can multiply successive rows of A by the successive diagonal entries of D, and to multiply A on the right by D, one can multiply successive columns of A by the successive diagonal entries of D.

Triangular Matrices

Definition

A square matrix in which all the entries above the main diagonal are zero is called ***lower triangular***, and a square matrix in which all the entries below the main diagonal are zero is called ***upper triangular***. A matrix that is either upper triangular or lower triangular is called ***triangular***.

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ 0 & a_{22} & a_{23} & a_{24} \\ 0 & 0 & a_{33} & a_{34} \\ 0 & 0 & 0 & a_{44} \end{bmatrix}$$

A general 4×4 upper triangular matrix

$$\begin{bmatrix} a_{11} & 0 & 0 & 0 \\ a_{21} & a_{22} & 0 & 0 \\ a_{31} & a_{32} & a_{33} & 0 \\ a_{41} & a_{42} & a_{43} & a_{44} \end{bmatrix}$$

A general 4×4 lower triangular matrix

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Properties of Triangular Matrices

Theorem

(a) The transpose of a lower triangular matrix is upper triangular, and the transpose of an upper triangular matrix is lower triangular.

(b) The product of lower triangular matrices is lower triangular, and the product of upper triangular matrices is upper triangular.

Example

Consider the upp

$$A=\begin{bmatrix}1 & 3 & -1\\ 0 & 2 & 4\\ 0 & 0 & 5\end{bmatrix},\quad B=\begin{bmatrix}3 & -2 & 2\\ 0 & 0 & -1\\ 0 & 0 & 1\end{bmatrix}$$

We can compute that both AB and BA are also upper triangular :

$$AB=\begin{bmatrix}3 & -2 & -2\\ 0 & 0 & 2\\ 0 & 0 & 5\end{bmatrix},\quad BA=\begin{bmatrix}3 & 5 & -1\\ 0 & 0 & -5\\ 0 & 0 & 5\end{bmatrix}$$

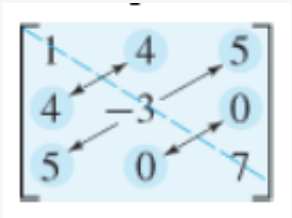
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Symmetric Matrices

Definition

A square matrix A is said to be symmetric if $A = A^T$.

It is easy to recognize a symmetric matrix by inspection: The entries on the main diagonal have no restrictions, but mirror images of entries *across* the main diagonal must be equal, as in this illustration :



Example :

The following matrices are symmetric, since each is equal to its own transpo

$$\begin{bmatrix} 7 & -3 \\ -3 & 5 \end{bmatrix}, \quad \begin{bmatrix} 1 & 4 & 5 \\ 4 & -3 & 0 \\ 5 & 0 & 7 \end{bmatrix}, \quad \begin{bmatrix} d_1 & 0 & 0 & 0 \\ 0 & d_2 & 0 & 0 \\ 0 & 0 & d_3 & 0 \\ 0 & 0 & 0 & d_4 \end{bmatrix}$$

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Properties of Symmetric Matrices

A square matrix A=[a_{ij}] is symmetric if and only if $(A)_{ij} = (A)_{ji}$, for all values of *i* and *j*. That is, $A^T = A$.

Theorem

If *A* and *B* are symmetric matrices with the same size, and if *k* is any scalar, then:

(a) A^T is symmetric.

(b) A + B and A – B are symmetric.

(c) k.A is symmetric.

(d) A.B is symmetric if and only if A.B = B.A.

(The product of two symmetric matrices is symmetric if and only if the matrices commute)

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INVERSE MATRIX AND ITS PROPERTIES

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Definition

If A is a square matrix, and if there exists a matrix B of the same size for which $AB = BA = I$, then A is said to be invertible (or nonsingular) and B is called an inverse of A. If no such matrix B exists, then A is said to be singular.

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Example

- Let matrix A dan B

$$A = \begin{pmatrix} 2 & -5 \\ -1 & 3 \end{pmatrix} \text{ dan } B = \begin{pmatrix} 3 & 5 \\ 1 & 2 \end{pmatrix}$$

Consider matrix multiplication AB and BA

$$AB = \begin{pmatrix} 2 & -5 \\ -1 & 3 \end{pmatrix}\begin{pmatrix} 3 & 5 \\ 1 & 2 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I \text{ (identity matrix)}$$

And

$$BA = \begin{pmatrix} 3 & 5 \\ 1 & 2 \end{pmatrix}\begin{pmatrix} 2 & -5 \\ -1 & 3 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I(\text{identity matrix})$$

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Inverse of 2 × 2 Matrix

Theorem

The matrix

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

is invertible if and only if $ad - bc \neq 0$,

in which case the inverse is given by the formula

$$A^{-1} = \frac{1}{ad-bc}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

Example: Calculating the inverse of the matrix $A = \begin{pmatrix} 6 & 1 \\ 5 & 2 \end{pmatrix}$

Solution :

$$A^{-1} = \frac{1}{7}\begin{pmatrix} 2 & -1 \\ -5 & 6 \end{pmatrix} = \begin{pmatrix} \frac{2}{7} & -\frac{2}{7} \\ -\frac{5}{7} & \frac{6}{7} \end{pmatrix}$$

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Theorem

1. If A and B are invertible matrices with the same size, then AB is invertible and
$(AB)^{-1} = B^{-1} A^{-1}$

2. If A is invertible and n is a nonnegative integer, then:
(a) A^{-1} is invertible and $(A^{-1})^{-1} = A$.
(b) An is invertible and $(A^{n})^{-1} = A^{-n} = (A^{-1})^{n}$.
(c) kA is invertible for any nonzero scalar k, and $(kA)^{-1} = k^{-1}A^{-1}$.

Inverse Technique : Inversion Algorithm

To find the inverse of an invertible matrix *A*, find a sequence of elementary row operations that reduces *A* to the identity and then perform that same sequence of operations on I_n to obtain A^{-1}.

To accomplish this we will adjoin the identity matrix to the right side of *A*, thereby producing a partitioned matrix of the form

$$[\ A \mid I\ ]$$

Then we will apply row operations to this matrix until the left side is reduced to *I*; these operations will convert the right side to A^{-1}, so the final matrix will have the form

$$[\ I \mid A^{-1}\ ]$$

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Example: Using Row Operations to Find A^{-1}.

Find the Inverse of

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 5 & 3 \\ 1 & 0 & 8 \end{pmatrix}$$

Solution :

- We want to reduce A to the identity matrix by row operations and simultaneously apply these operations to I to produce A^{-1}.

For this we will adjoin the identity matrix to the right side of A, thereby producing a partitioned matrix of the form $[A \mid I]$

- Then we will apply row operations to this matrix until the left side is reduced to I; these operations will convert the right side to A^{-1}, so the final matrix will have the form $[I \mid A^{-1}]$

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Example

The computations are as follows:

$$\left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 2 & 5 & 3 & 0 & 1 & 0 \\ 1 & 0 & 8 & 0 & 0 & 1 \end{array}\right]$$

$$\left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & 1 & -3 & -2 & 1 & 0 \\ 0 & -2 & 5 & -1 & 0 & 1 \end{array}\right]$$

← We added −2 times the fir row to the second and −1 the first row to the third.

$$\left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & 1 & -3 & -2 & 1 & 0 \\ 0 & 0 & -1 & -5 & 2 & 1 \end{array}\right]$$

← We added 2 times the second row to the third.

$$\left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & 1 & -3 & -2 & 1 & 0 \\ 0 & 0 & 1 & 5 & -2 & -1 \end{array}\right]$$

← We multiplied the third row by −1.

$$\left[\begin{array}{ccc|ccc} 1 & 2 & 0 & -14 & 6 & 3 \\ 0 & 1 & 0 & 13 & -5 & -3 \\ 0 & 0 & 1 & 5 & -2 & -1 \end{array}\right]$$

← We added 3 times the third row to the second and −3 times the third row to the first.

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -40 & 16 & 9 \\ 0 & 1 & 0 & 13 & -5 & -3 \\ 0 & 0 & 1 & 5 & -2 & -1 \end{array}\right]$$

← We added −2 times the second row to the first.

Thus,

$$A^{-1} = \left[\begin{array}{ccc} -40 & 16 & 9 \\ 13 & -5 & -3 \\ 5 & -2 & -1 \end{array}\right]$$

Inverse of a Product

Theorem

If A and B are invertible matrices with the same size, then $A.B$ is invertible and

$$(A.B)^{-1} = B^{-1}.A^{-1}$$

The theorem above can be extended to three or more matrices :

A product of any number of invertible matrices is invertible, and the inverse of the product is the product of the inverses in the reverse order.

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More Properties of Inverse Matrix

Here is some properties of inverse matrix related to exponent and transpose.

Theorem

If A is invertible and n is a nonnegative integer, then:

(a) A^{-1} is invertible and $(A^{-1})^{-1} = A$.

(b) A^{n} is invertible and $(A^{n})^{-1} = A^{-n} = (A^{-1})^{n}$.

(c) k.A is invertible for any nonzero scalar k, and $(k.A)^{-1} = k^{-1}.A^{-1}$.

(a) A^{T} is invertible and $(A^{T})^{-1} = (A^{-1})^{T}$.

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Solving Linear Systems by Matrix Inversion

So far we have studied two *procedures* for solving linear system: Gauss–Jordan elimination and Gaussian elimination. The following theorem provides an actual *formula* for the solution of a linear system of *n* equations in *n* unknowns in the case where the coefficient matrix is invertible.

Theorem

If A is an invertible *n* x *n* matrix, then for each *n* x 1 matrix **b**, the system of equations $A.\boldsymbol{x} = \boldsymbol{b}$ has exactly one solution, namely

$$\boldsymbol{x} = A^{-1}.\boldsymbol{b}.$$

The theorem above can be proved easily by multiplying each both sides by A^{-1} from the left.

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Example

Consider the system of linear equations

$$\begin{aligned} x_1 + 2x_2 + 3x_3 &= 5 \\ 2x_1 + 5x_2 + 3x_3 &= 3 \\ x_1 \qquad\quad + 8x_3 &= 17 \end{aligned}$$

In matrix form this system can be written as A.**x** = **b**, where

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 3 \\ 1 & 0 & 8 \end{bmatrix}, \quad \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 5 \\ 3 \\ 17 \end{bmatrix}$$

Example (Cont.)

In previous example we showed that A is invertible and

$$A^{-1} = \begin{bmatrix} -40 & 16 & 9 \\ 13 & -5 & -3 \\ 5 & -2 & -1 \end{bmatrix}$$

The solution of the system is

$$\mathbf{x} = A^{-1}\mathbf{b} = \begin{bmatrix} -40 & 16 & 9 \\ 13 & -5 & -3 \\ 5 & -2 & -1 \end{bmatrix} \begin{bmatrix} 5 \\ 3 \\ 17 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix}$$

or $x_1 = 1$, $x_2 = -1$, $x_3 = 2$.

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Determinant

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Determinant Matrix 2×2

Let A be a 2×2 $matrix$

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

The matrix *A* is invertible if and only if $ad - bc \neq 0$ and that the expression $ad - bc$ is called **determinant** of the matrix A. Recall also that this determinant is denoted by writing

$$\det(A) = ad - bc \text{ or } \begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$$

$Note$: $\det(A)$ is a number, whereas A is a matrix.

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Inverse and determinant

Let A be an $n \; x \; n$ matrix

.

The matrix A is invertible if and only $\det(A) \neq 0$.

Calculating Determinants and Inverse by Cofactor Ekxpansion

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Minor and Cofactor

Definition

If A is a square matrix, then the **minor** of entry a_{ij} is denoted by M_{ij} and is defined to be the determinant of the submatrix that remains after the i-th row and j-th column are deleted from A.

The number $(-1)^{i+j} M_{ij}$ is denoted by C_{ij} and is called the **cofactor** of the entry a_{ij}.

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Minor and Cofactor

Definition

If A is a square matrix, then the minor of entry a_{ij} is denoted by M_{ij} and is defined to be the determinant of the submatrix that remains after the i-th row and j-th column are deleted from A. The number $(-1)^{i+j} M_{ij}$ is denoted by C_{ij} and is called the cofactor of entry a_{ij}

Example : Finding Minors and Cofactors

$$\boldsymbol{A} = \begin{pmatrix} 3 & 1 & -4 \\ 2 & 5 & 6 \\ 1 & 4 & 8 \end{pmatrix}$$

Solution

The minor and cofactor of entry a_{ij} is

$$\boldsymbol{M_{11}} = \begin{vmatrix} \boldsymbol{3} & \boldsymbol{1} & \boldsymbol{-4} \\ \boldsymbol{2} & \boldsymbol{5} & \boldsymbol{6} \\ \boldsymbol{1} & \boldsymbol{4} & \boldsymbol{8} \end{vmatrix} = \begin{vmatrix} \boldsymbol{5} & \boldsymbol{6} \\ \boldsymbol{4} & \boldsymbol{8} \end{vmatrix} = \boldsymbol{16}$$

$$\boldsymbol{C_{11}} = -1^{1+1}\boldsymbol{M_{11}} = \boldsymbol{M_{11}} = \boldsymbol{16}$$

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Determinants by Cofactor Expansion

Definition

If A is an $n \times n$ matrix, then the number obtained by multiplying the entries in any row or column of A by the corresponding cofactors and adding the resulting products is called the determinant of A, and the sums themselves are called cofactor expansions of A.

$$\det \mathrm{A} = \boldsymbol{a_{1j}C_{1j}} + \boldsymbol{a_{2j}C_{2j}} + \cdots + \boldsymbol{a_{nj}C_{nj}}$$

(Cofactor Expansion along the jth column**)**

$$\det A = \boldsymbol{a_{i1}C_{i1}} + \boldsymbol{a_{i2}C_{i2}} + \cdots + \boldsymbol{a_{in}C_{in}}$$

(Cofactor Expansion along the ith row**)**

Example : Find the determinant of the matrix by cofactor expansion along the first row.

$$\boldsymbol{A} = \begin{pmatrix} 3 & 1 & -4 \\ 2 & 5 & 6 \\ 1 & 4 & 8 \end{pmatrix}$$

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Determinants by Cofactor Expansion

Example : Find the determinant of the matrix by cofactor expansion along the first row.

$$\boldsymbol{A} = \begin{pmatrix} 3 & 1 & -4 \\ 2 & 5 & 6 \\ 1 & 4 & 8 \end{pmatrix}$$

Solution

The minor and cofactor of entry a_{ij} is

$$\boldsymbol{det(A)} = \begin{vmatrix} \mathbf{3} & \mathbf{1} & \mathbf{-4} \\ \mathbf{2} & \mathbf{5} & \mathbf{6} \\ \mathbf{1} & \mathbf{4} & \mathbf{8} \end{vmatrix} = \mathbf{3}\begin{vmatrix} \mathbf{5} & \mathbf{6} \\ \mathbf{4} & \mathbf{8} \end{vmatrix} - \mathbf{1}\begin{vmatrix} \mathbf{2} & \mathbf{6} \\ \mathbf{1} & \mathbf{8} \end{vmatrix} - \mathbf{4}\begin{vmatrix} \mathbf{2} & \mathbf{5} \\ \mathbf{1} & \mathbf{4} \end{vmatrix}$$

$$= \mathbf{3(16) - 10 - 4(3)} = \mathbf{-1}$$

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2. Inverse of a Matrix Using Its Adjoint

Definition : Adjoint

If A is any $n \times n$ matrix and C_{ij} is the cofactor of a_{ij}, then the matrix

$$\begin{bmatrix} C_{11} & C_{12} & \cdots & C_{1n} \\ C_{21} & C_{22} & \cdots & C_{2n} \\ \vdots & \vdots & & \vdots \\ C_{n1} & C_{n2} & \cdots & C_{nn} \end{bmatrix}$$

is called the ***matrix of cofactors from A***. The transpose of this matrix is called the ***adjoint of A*** and is denoted by adj(A).

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Inverse of a Matrix Using Its Adjoint

If A is an invertible matrix, then

$$\mathbf{A^{-1}} = \frac{\mathbf{1}}{\mathbf{det(A)}}\ \mathbf{adj}\ \mathbf{(A)}$$

Example : Calculating the Inverse of matrix

$$\mathrm{A} = \begin{pmatrix} 3 & 2 & -1 \\ 1 & 6 & 3 \\ 2 & -4 & 0 \end{pmatrix}$$

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Inverse of a Matrix Using Its Adjoint

Solution :

a. Find the Determinant of A

$$\begin{vmatrix} 3 & 2 & -1 \\ 1 & 6 & 3 \\ 2 & -4 & 0 \end{vmatrix} = 64$$

b. Find the adjoin of matrix A

$$\mathbf{Adj}\ (\mathrm{A}) = \begin{pmatrix} 12 & 4 & 12 \\ 6 & 2 & -10 \\ -16 & 16 & 16 \end{pmatrix}$$

$$\mathbf{A^{-1}} = \frac{\mathbf{1}}{\mathbf{det(A)}}\ \mathbf{adj}\ (\mathbf{A})$$

$$\mathbf{A^{-1}} = \frac{\mathbf{1}}{\mathbf{64}} \begin{pmatrix} 12 & 4 & 12 \\ 6 & 2 & -10 \\ -16 & 16 & 16 \end{pmatrix} = \begin{pmatrix} \frac{\mathbf{12}}{\mathbf{64}} & \frac{\mathbf{4}}{\mathbf{64}} & \frac{\mathbf{12}}{\mathbf{64}} \\ \frac{\mathbf{6}}{\mathbf{64}} & \frac{\mathbf{2}}{\mathbf{64}} & -\frac{\mathbf{10}}{\mathbf{64}} \\ -\frac{\mathbf{16}}{\mathbf{64}} & \frac{\mathbf{16}}{\mathbf{64}} & \frac{\mathbf{16}}{\mathbf{64}} \end{pmatrix}$$

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Properties of determinant

Properties of Determinants

Theorem

Let A be a square matrix.

(a) If A has a row of zeros or a column of zeros, then $\det(A)=0$.

(b) $\det(A) = \det(A^T)$.

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Properties of Determinants

Theorem

Suppose that A and B are n x n matrices and k is any scalar. We have the following properties :

(a) $\det(A.B) = \det(A).\det(B)$

(b) $\det(k.A) = k^n.\det(A)$

(c) In general, $\det(A+B) \neq \det(A)+\det(B)$

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Determinants of Triangular Matrices

Theorem

If *A* is an *n* x *n* triangular matrix (upper triangular, lower triangular, or diagonal),

then det(A) is the product of the entries on the main diagonal of the matrix; that is, det(A) = $a_{11}.a_{22}.....a_{nn}$.

$$\begin{vmatrix} a_{11} & 0 & 0 & 0 \\ a_{21} & a_{22} & 0 & 0 \\ a_{31} & a_{32} & a_{33} & 0 \\ a_{41} & a_{42} & a_{43} & a_{44} \end{vmatrix} = a_{11} \begin{vmatrix} a_{22} & 0 & 0 \\ a_{32} & a_{33} & 0 \\ a_{42} & a_{43} & a_{44} \end{vmatrix}$$

$$= a_{11}a_{22} \begin{vmatrix} a_{33} & 0 \\ a_{43} & a_{44} \end{vmatrix}$$

$$= a_{11}a_{22}a_{33}|a_{44}| = a_{11}a_{22}a_{33}a_{44}$$

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Calculating Determinants by Row Reduction

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Theorem

Let A be an $n \times n$ matrix.

(a) If B is the matrix that results when a single row or single column of A is multiplied by a scalar k, then $\det(B) = k \ \det(A)$

(b) If B is the matrix that results when two rows or two columns of A are interchanged,
then $\det(B) = -\det(A)$.

(c) If B is the matrix that results when a multiple of one row of A is added to another row or when a multiple of one column is added to another column, then $\det(B) = \det(A)$.

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Relationship	Operation
$\begin{vmatrix} ka_{11} & ka_{12} & ka_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} = k \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$ $\det(B) = k\det(A)$	The first row of A is multiplied by k.
$\begin{vmatrix} a_{21} & a_{22} & a_{23} \\ a_{11} & a_{12} & a_{13} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} = - \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$ $\det(B) = -\det(A)$	The first and second rows of A are interchanged.
$\begin{vmatrix} a_{11} + ka_{21} & a_{12} + ka_{22} & a_{13} + ka_{23} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$ $\det(B) = \det(A)$	A multiple of the second row of A is added to the first row.

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Example

Find the determinant of $\begin{pmatrix} 0 & 1 & 5 \\ 3 & -6 & 9 \\ 2 & 6 & 1 \end{pmatrix}$

Solution

$$\det(A) = \begin{vmatrix} 0 & 1 & 5 \\ 3 & -6 & 9 \\ 2 & 6 & 1 \end{vmatrix} = - \begin{vmatrix} 3 & -6 & 9 \\ 0 & 1 & 5 \\ 2 & 6 & 1 \end{vmatrix}$$

← The first and second rows of A where interchanged.

$$= -3 \begin{vmatrix} 1 & -2 & 3 \\ 0 & 1 & 5 \\ 2 & 6 & 1 \end{vmatrix}$$

← A common factor of 3 from the first row was taken through the determinant sign.

$$= -3 \begin{vmatrix} 1 & -2 & 3 \\ 0 & 1 & 5 \\ 0 & 10 & -5 \end{vmatrix}$$

← −2 times the first row was added to the third row.

$$= -3 \begin{vmatrix} 1 & -2 & 3 \\ 0 & 1 & 5 \\ 0 & 0 & -55 \end{vmatrix}$$

← −10 times the second row was added to the third row.

$$= (-3)(-55) \begin{vmatrix} 1 & -2 & 3 \\ 0 & 1 & 5 \\ 0 & 0 & 1 \end{vmatrix}$$

← A common factor of −55 from the last row was taken through the determinant sign.

$$= (-3)(-55)(1) = 165$$

The Cramer's Rule

Cramer's Rule

Our next theorem uses the formula for the inverse of an invertible matrix to produce a formula, called ***Cramer's rule***, for the solution of a linear system $A.\mathbf{x} = \mathbf{b}$ of n equations in n unknowns in the case where the coefficient matrix A is invertible (or, equivalently, when $\det(A) \neq 0$).



Gabriel Cramer (1704–1752)

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Cramer's Rule

Theorem (Cramer's Rule)

If $A.\mathbf{x}=\mathbf{b}$ is a system of n linear equations in n unknowns such that $\det(A) \neq 0$, then the system has a unique solution. This solution is

$$x_1 = \frac{\det(A_1)}{\det(A)}, \quad x_2 = \frac{\det(A_2)}{\det(A)}, \ldots, \quad x_n = \frac{\det(A_n)}{\det(A)}$$

where A_j is the matrix obtained by replacing the entries in the j-th column of A by the entries in the matrix

$$\mathbf{b} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix}$$

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Example

Use Cramer's rule to solve

$$\begin{aligned} x_1 &+ &&+ 2x_3 &= 6 \\ -3x_1 &+ 4x_2 &+ 6x_3 &= 30 \\ -x_1 &- 2x_2 &+ 3x_3 &= 8 \end{aligned}$$

$$A = \begin{bmatrix} 1 & 0 & 2 \\ -3 & 4 & 6 \\ -1 & -2 & 3 \end{bmatrix}, \quad A_1 = \begin{bmatrix} 6 & 0 & 2 \\ 30 & 4 & 6 \\ 8 & -2 & 3 \end{bmatrix},$$

$$A_2 = \begin{bmatrix} 1 & 6 & 2 \\ -3 & 30 & 6 \\ -1 & 8 & 3 \end{bmatrix}, \quad A_3 = \begin{bmatrix} 1 & 0 & 6 \\ -3 & 4 & 30 \\ -1 & -2 & 8 \end{bmatrix}$$

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Example

Therefore,

$$cx_1 = \frac{\det(A_1)}{\det(A)} = \frac{-40}{44} = \frac{-10}{11}, \quad x_2 = \frac{\det(A_2)}{\det(A)} = \frac{72}{44} = \frac{18}{11},$$

$$x_3 = \frac{\det(A_3)}{\det(A)} = \frac{152}{44} = \frac{38}{11}$$

Note : For $n > 3$, it is usually more efficient to solve a linear system with n equations in n unknowns by Gauss–Jordan elimination than by Cramer's rule. Its main use is for obtaining properties of solutions of a linear system without actually solving the system.

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References

Howard Anton and Anton Kaul. (2019). Elementary Linear Algebra. 12. Wiley. New Jersey. ISBN: 978-1-119-40677-8

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Thank You

